

Noise-Aware ANNNI Phase Diagnostics

Low-CNOT preparation, error mitigation, and their limits

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Problem and physical protocol

We study the periodic axial next-nearest-neighbor Ising chain $H = -\sum ZZ(nn) + \kappa \sum ZZ(nnn) - h \sum X$, $J_1=1$. Competition between interactions makes low-field preparation and phase-feature interpretation difficult. We separate state accuracy, observable accuracy, and phase-diagram evidence. $N=8$ circuits cover 420 positive-field coordinates; $N=12$ and large-OBC MPS calculations supply qualified extensions.

Preparation, noise and diagnostics

B0 is a fixed six-layer Hamiltonian variational ansatz; B3 adaptively grows energy-optimized Pauli rotations from multiple references and selects using a frozen energy/resource rule. Every executed CNOT receives target-only $D_p(\rho)=(1-p)\rho+p/3(X\rho X+Y\rho Y+Z\rho Z)$, including reference preparation and folding. Explicit gates retain wire-0-MSB order and recorded connectivity. Complete $C(r)$, all discrete structure factors and M_x are measured. Frozen D3 is a PCA/cluster resemblance diagnostic, not an independent physical truth oracle.

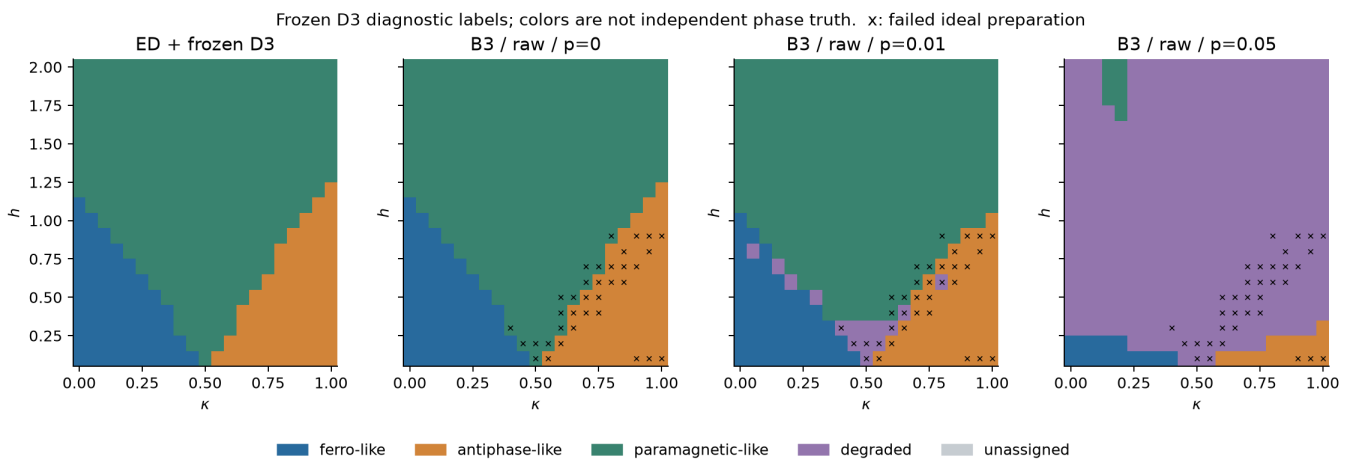
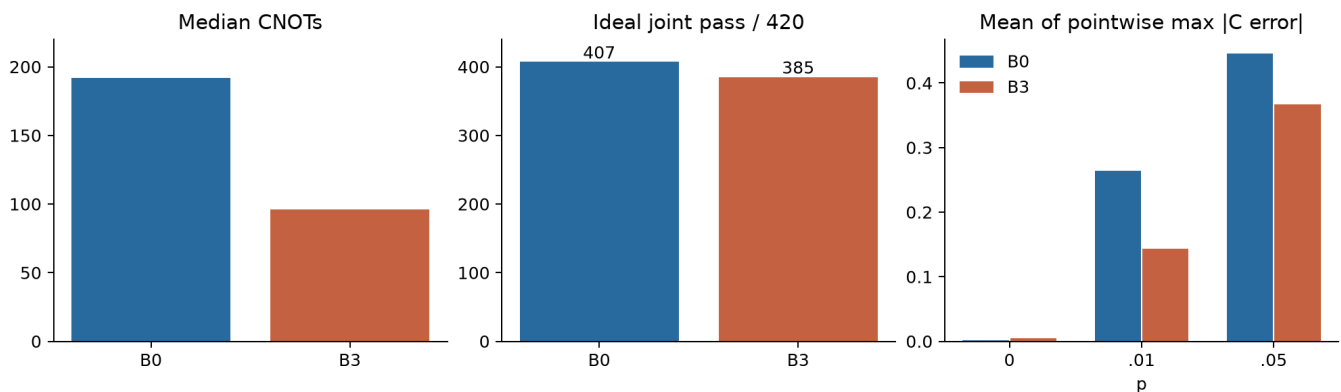


Figure 1. N8 PBC, $h>0$. ED and three raw B3 noise levels share frozen D3 and color scale. Colors are phase-like estimates. Crosses retain preparation failures; reference coverage is separately available in the notebook.

The low-CNOT tradeoff

B0 to B3 reduces median CNOTs from 192 to 96. Ideal joint preparation coverage decreases from 407/420 to 385/420. At $p=.01$, mean pointwise maximum correlation error decreases from 0.2638 to 0.1432; B3 improves 402/420 coordinates. These are correlation errors, not phase-classification errors. Joint acceptance keeps $\delta e \leq .001$ per spin, $C/SF/M_x$ errors $\leq .02$ and squared ED fidelity $\geq .99$.



Genuine equal-CNOT-shots comparison

On 45 fixed coordinates we compare raw, quadratic zero-noise extrapolation (ZNE), and global-X symmetry verification (SV), with 32 measurement repeats. $G = \text{sum}(\text{shots} \times \text{literal CNOTs})$ is matched per coordinate: raw/SV use 100,000 shots; ZNE uses 33,334 across folds 1/3/5, with weights 15/8,-5/4,3/8. SV estimates $(O+OP)/(1+P)$ from all required signed joint settings. Settings and one-qubit costs remain distinct; equal G is not equal total execution cost.

Observable result and negative control

ED-target MSE averages the unchanged 17 components (eight C, eight Fourier-related SF, Mx). At $p=.01$, raw/ZNE/SV give 0.01754430/0.00162271/0.01235293. At $p=.05$ they give 0.10511371/0.07863611/0.10231166. ZNE reduces mean MSE by 90.75% and 25.19%, improving coordinate-mean errors at all 45 points. At $p=0$ it worsens all 45. Bias reduction is accompanied by larger sampling MSE. Only six points have independent interior labels; these results do not establish 45-point phase accuracy.

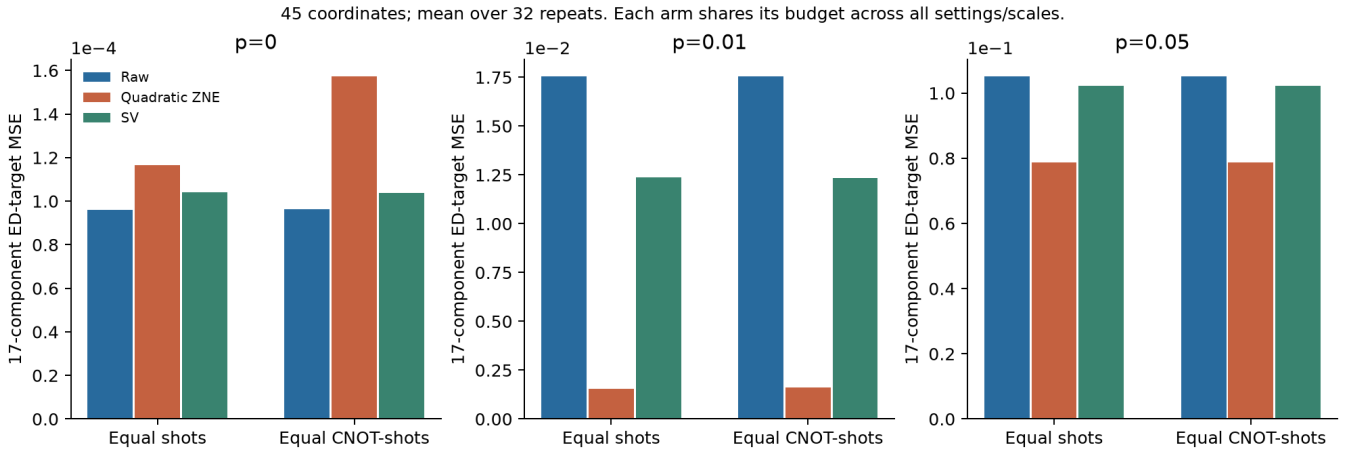


Figure 2. All 45 coordinates retained. Each value averages 32 repeats; related C/SF components retain their historical equal weighting. Equal-shots and equal-CNOT-shots are distinct experiments.

Why feature reconstruction remains limited

All six 17-point windows at $\kappa=0, .3, .45, .5, .55, .8$ are executed, with identical grids and frozen matching. The .5 reference fails the frozen peak-selection criterion. The other five supply finite-size RN features, not thermodynamic critical points. At $p=.05$ and 100k equal shots, raw matches 2.96875/5 window equivalents across repeats; ZNE 2.875/5. Endpoint domination, competing peaks, preparation error and branch sensitivity remain explicit. Observable MSE and frozen feature matching quantify distinct outcomes; these matches describe finite-size response features.

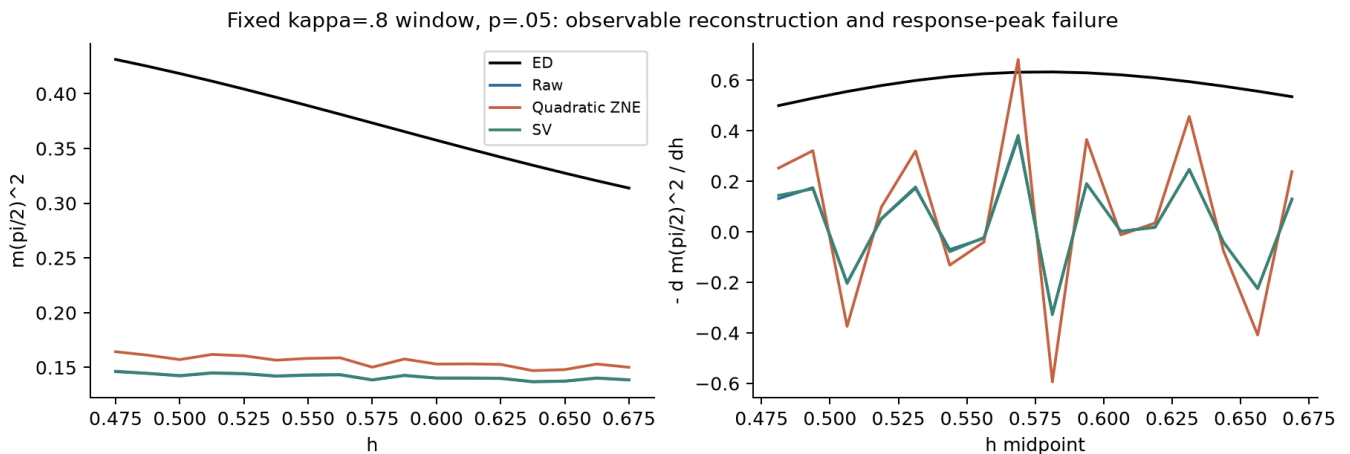


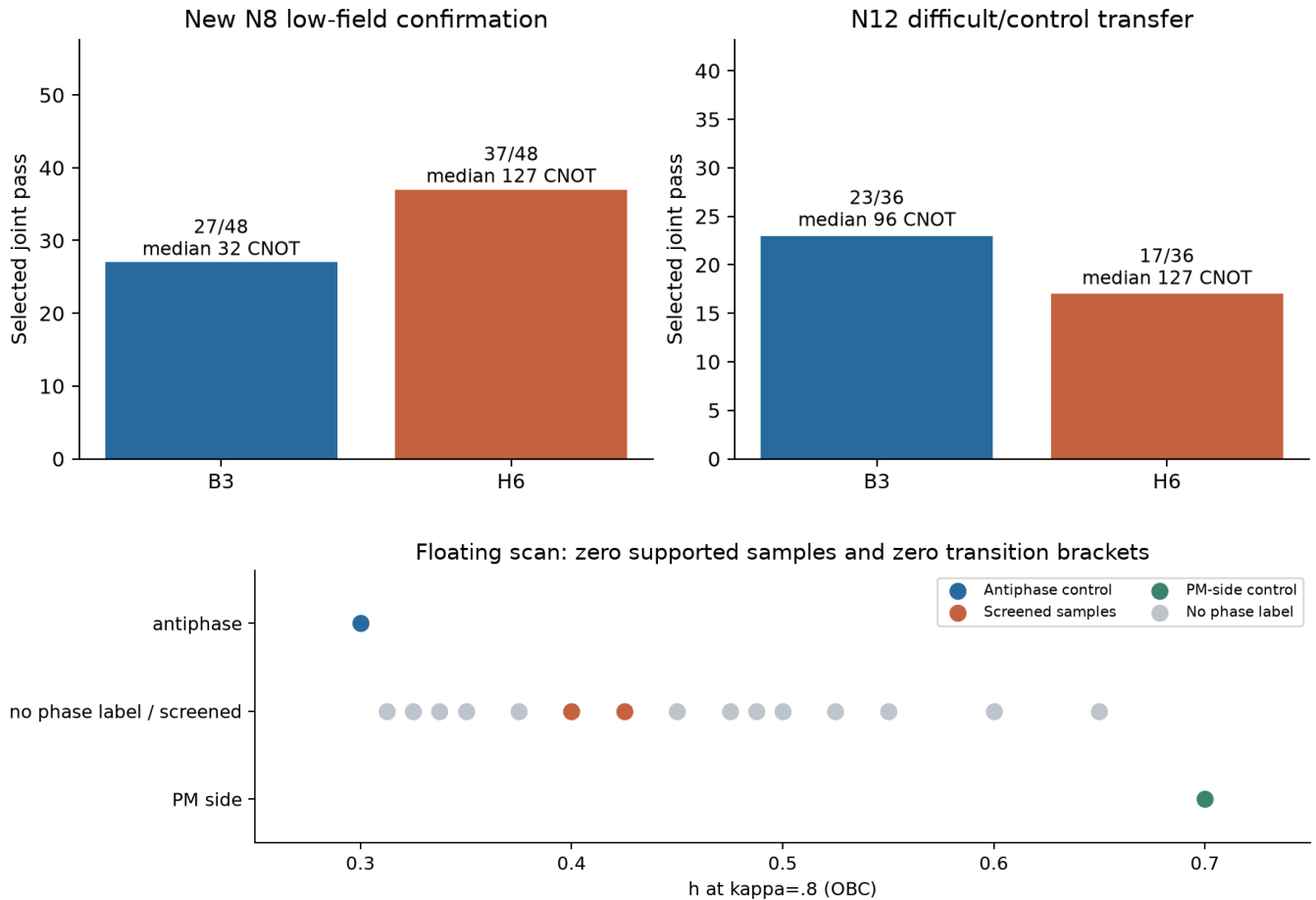
Figure 3. The prespecified $\kappa=.8$ window at $p=.05$; exact expectations isolate noise/preparation effects. All endpoints and both competing response features are retained. No posthoc nearest-peak selection.

Low-field construction and transfer

The frozen H6 union of physical/domain-wall candidates increases candidate availability from 27/48 to 45/48, and selected joint passes from 27/48 to 37/48. Median CNOTs rise from 32 to 127; objective calls rise by 7.45x. This is not an equal-cost win. N12 difficult/control transfer gives B3 23/36 versus H6 17/36. H6 is the physical/domain-wall union; it improves the N8 low-field cohort and reduces passes on this N12 cohort.

Floating search and other extensions

A controlled $\kappa=0.8$ OBC slice supports an antiphase sample at $h=0.3$ and a paramagnetic-side sample at $h=0.7$. At $h=0.425$, size, entropy and Friedel diagnostics disagree. The final evidence contains zero supported floating samples and zero established transition brackets. Bond-dimension convergence tests numerical stability; gap and correlation-length scaling determine phase identification. A $\chi=512$ timeout is a resource limit, not physical exclusion. Archived dynamics quantify Trotter/noise effects; multiple detectors and N12 experiments remain in the notebook. The published evidence is frozen.



Conclusion and reproducibility

The evidence supports lower-gate preparation and cost-controlled mitigation as tools for observable reconstruction, while preserving the gap to reliable phase-boundary inference. R0 region interiors, RN same-size diagnostics, and Rlarge size-qualified evidence are separate. Main figures and statistics rebuild from bundled numerical data and joint counts. Lightweight explicit-gate and SDK checks are supplied. V1 deterministic saved-data verification, V2 physical checks and environment-gated V3 exact random replay are reported independently.

References: Q-SITE 2026 Scientific Track, snapshot 57f9a537 (local upstream/); Tang et al., qubit-ADAPT-VQE, arXiv:1911.10205; ANNNI tensor-network study, arXiv:2402.11022; floating-phase DMRG, arXiv:cond-mat/0702676; symmetry verification, arXiv:1807.10050. Published XX/Z conventions are related to ZZ/X by local Hadamards; boundaries and sizes are not interchangeable.